

$$\Psi = \int e^{\frac{i}{\hbar} \int \left(\frac{R}{16\pi G} - \frac{1}{4} F^2 + \bar{\psi} i \not{D} \psi - \lambda \phi \bar{\psi} \psi + |D\phi|^2 - V(\phi) \right)}$$

path integral Feynmann
 imaginary unit
 spacetime-relativity Einstein
 strong/weak/e.m. interactions Maxwell Yang-Mills
 $\phi - \psi$ interaction Yukawa
 Schrödinger wave function
 Euler exponential
 Planck quantum
 Newton gravitation
 Dirac relativistic wave function
 Kobayashi-Maskawa CKM matrix
 Higgs Boson

Molecule Spectroscopy



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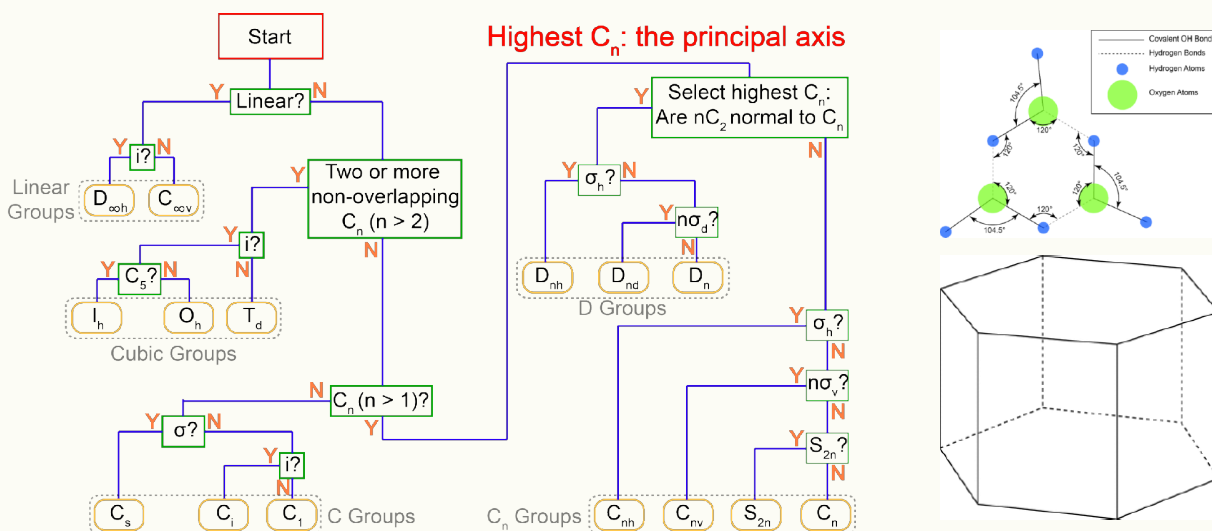
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Assignment #1

Problem 1.1. Please analyze the symmetry of the following shapes by 1. listing **all the symmetry elements**; 2. using the flow chart to determine which point groups these structures belong to.



- (a) A **planar** 'cluster' of a tri-water-molecule 'bonded' with hydrogen bonds between oxygen and hydrogen atoms (see the figure below for detailed structural parameters). **(0.5 point)**
- (b) Hexagonal Prism (It has two identical parallel hexagonal bases-top and bottom faces, and six rectangular faces connecting corresponding edges of the hexagons). **(0.5 point)**

Solution.

- (a) i. Symmetry elements.
1. Identity (E).
 2. Principal axis (C_3): An axis through the center of the hexagon perpendicular to the plane. Rotating the cluster by $360^\circ/3$ maps each water molecule onto the next one.
 3. Horizontal mirror plane (σ_h): Since the cluster is planar, the plane containing all the atoms acts as a mirror plane.
 4. Improper rotation axis (S_3): Resulting from performing a C_3 rotation followed by a reflection in the σ_h plane.
- ii. Mapping the symmetry elements to the flow chart.
1. Nonlinear molecule.
 2. Only have one C_3 axis.
 3. C_n ($n = 3 > 1$) satisfies.
 4. No C_2 axis.
 5. Horizontal mirror plane σ_h exists.

Linear (N) $\gg$ ≥ 2 non-overlapping $C_{n>2}$ (N) $\gg$ C_n ($n > 1$) (Y) $\gg$ nC_2 normal to highest C_n (N) $\gg$ σ_h (Y) $\gg$ C_{3h}

- (b) i. Symmetry elements.
1. Identity (E).
 2. Principal axis (C_6): A 6-fold rotation axis passing through the centers of the top and bottom hexagonal faces, *coincident with* C_3 and C_2 rotations.
 3. Perpendicular C_2 axes ($6 \times C_2$): Three pass through the centers of opposite vertical rectangular faces, and three pass through opposite vertical edges.
 4. One horizontal mirror plane and six vertical mirror planes: (σ_h , $3 \times \sigma_v$, $3 \times \sigma_d$).
 5. Inversion center (i).
 6. Improper rotation axes (S_6 , S_3 , S_2).
- ii. Mapping the symmetry elements to the flow chart.
1. Nonlinear molecule.
 2. No $C_{n>2}$ axes perpendicular to the principal axis: Only have C_2 perpendicular to it.
 3. C_n ($n = 6 > 1$) satisfies.
 4. $6C_2$ axes perpendicular to C_6 axis.
 5. Horizontal mirror plane σ_h exists.

Linear (N) $\gg \geq 2$ non-overlapping $C_{n>2}$ (N) $\gg C_n$ ($n > 1$) (Y) $\gg nC_2$ normal to highest C_n (Y) $\gg \sigma_h$ (Y) $\gg D_{6h}$

Problem 1.2. Continue working on the decomposition of the reducible representation of BF_3 into irreducible representations (irreps., as various molecular modes).

- (a) Point out which irreps. correspond to 1. translational, 2. rotational, and 3. vibrational modes. **(0.5 point)**

Remark (The BF_3 (D_{3h}) point group character table).

D_{3h}	E	$2C_3(z)$	$3C'_2$	$\sigma_h(xy)$	$2S_3$	$3\sigma_v$	Linear functions, rotations	Quadratic functions
A'_1	+1	+1	+1	+1	+1	+1	—	$x^2 + y^2, z^2$
A'_2	+1	+1	−1	+1	+1	−1	R_z	—
E'	+2	−1	0	+2	−1	0	(x, y)	$x^2 - y^2, xy$
A''_1	+1	+1	+1	−1	−1	−1	—	—
A''_2	+1	+1	−1	−1	−1	+1	z	—
E''	+2	−1	0	−2	+1	0	(R_x, R_y)	(xz, yz)

- (b) BF_3 has several vibrational modes. Please prove step by step why the symmetric stretching mode is A'_1 mode. **(2 points)**
- (c) For all vibrational modes for BF_3 , point out their IR and Raman activity. **(0.5 point)**

Solution.

- (a) i. Translational modes: $\Gamma_{\text{trans}} = A_2'' + E'$.
 ii. Rotational Modes: $\Gamma_{\text{rot}} = A_2' + E''$.
 iii. Vibrational Modes: Calculate Γ_{tot} first. The traces of the operators are

$$\chi(E) = 1 + 1 + 1 = 3, \quad \chi(C_3) = 2 \cos(120^\circ) + 1 = 0, \quad \chi(C_2') = 1 - 1 - 1 = -1, \\ \chi(\sigma_h) = 1 + 1 - 1 = 1, \quad \chi(S_3) = 2 \cos(120^\circ) - 1 = -2, \quad \chi(\sigma_v) = 1 - 1 + 1 = 1.$$

Then, $\Gamma_{\text{tot}} = \frac{1}{h} \sum n_R \chi_R \chi_i = \{12, 0, -2, 4, -2, 2\} = A_1' + A_2' + 3E' + 2A_2'' + E''$. The irreps. corresponding to vibrational mode is

$$\Gamma_{\text{vib}} = \Gamma_{\text{tot}} - \Gamma_{\text{trans}} - \Gamma_{\text{rot}} = A_1' + 2E' + A_2''.$$

- (b) *Proof.* # of bonds remain unmoved under the six operations are

Operation	$1E$	$2C_3$	$3C_2'$	$1\sigma_h$	$2S_3$	$3\sigma_v$
N_{unmoved}	3	0	1	3	0	1

Substituting into the reduction formula $n_i = \frac{1}{h} \sum n_R \chi(R) \chi_i(R)$

$$n(A_1') = \frac{1}{12} [(3)(1)(1) + (0)(2)(+1) + (1)(3)(1) + (3)(1)(1) + (0)(2)(+1) + (1)(3)(1)] = 1, \\ n(E') = \frac{1}{12} [(3)(1)(2) + (0)(2)(-1) + (1)(3)(0) + (3)(1)(2) + (0)(2)(-1) + (1)(3)(0)] = 1,$$

while other irreps. yield 0. Therefore, $\Gamma_{\text{stretch}} = A_1' + E'$. □

- (c) Due to the requirements of IR active and Raman active

- i. IR Active: The mode must transform as the linear functions (x , y , or z), as this indicates a change in the molecular dipole moment.
 ii. Raman Active: The mode must transform as the quadratic functions (x^2 , y^2 , z^2 , xy , xz , yz), as this indicates a change in the molecular polarizability.

Evaluate the three modes from $\Gamma_{\text{vib}} = A_1' + 2E' + A_2''$ by comparing with D_{3h} group character table

- i. A_1' : IR inactive, Raman active. iii. E' : IR active, Raman active.
 ii. A_2'' : IR active, Raman inactive.

Problem 1.3. $W(\text{CO})_6$ (O_h point group) has three degenerate IR active vibrational modes with irreducible representation T_{1u} , as shown in the partial character table below

O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2 = (C_4)^2$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$	Linear functions, rotations	Quadratic functions
T_{1u}	+3	0	-1	+1	-1	-3	-1	0	+1	+1	(x, y, z)	—

Please follow the same following procedure and prove the character of $3C_2$ (3 C_2 -axes along x , y , and z axes) is -1 for T_{1u} irreps. (1 point)

Procedure: C_3 about a body diagonal (111)

A representative 120° rotation gives

$$x \rightarrow y, \quad y \rightarrow z, \quad z \rightarrow x.$$

So

$$T_x \rightarrow T_y, \quad T_y \rightarrow T_z, \quad T_z \rightarrow T_x.$$

Matrix:

$$D^{(T_{1u})}(C_3) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Character:

$$\chi(C_3) = 0.$$

So the $8C_3$ class has character 0.

Remark. The transformation of T_x, T_y, T_z after $C_2(z)$ operation, the corresponding matrix and the trace (the final value of character) are required.

Solution. A $C_2(z)$ operation represents a 180° rotation about the z -axis, which gives

$$x \rightarrow -x, \quad y \rightarrow -y, \quad z \rightarrow z.$$

Consequently, the translational basis functions (T_x, T_y, T_z) transform as:

$$T_x \rightarrow -T_x, \quad T_y \rightarrow -T_y, \quad T_z \rightarrow T_z.$$

The transformation can be represented by the matrix

$$D^{(T_{1u})}(C_2) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The character χ is defined as the trace of the representative matrix

$$\chi(C_2) = \text{Tr}(D^{(T_{1u})}(C_2)) = (-1) + (-1) + 1 = -1.$$

So, the $3C_2$ class in the O_h point group has a character of -1 for the T_{1u} irreducible representation.